RQ Answers

omitted

2024-04-05

Data overview

```
glimpse(all_data)
```

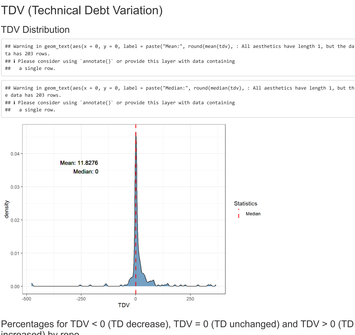
```
## Rows: 16,383
## Columns: 11
## $ repo                <chr> "helix", "helix", "helix", "helix", "helix", "heli…
## $ pr_number           <int> 1449, 1449, 1449, 1449, 1449, 1449, 1449, 1449, 14…
## $ rule                <chr> "java:S116", "java:S116", "java:S106", "java:S1074…
## $ severity            <chr> "MINOR", "MINOR", "MAJOR", "MINOR", "MINOR", "CRIT…
## $ file                <chr> "helix-core/src/test/java/org/apache/helix/integra…
## $ type                <chr> "CODE_SMELL", "CODE_SMELL", "CODE_SMELL", "CODE_SM…
## $ status              <chr> "OPEN", "OPEN", "OPEN", "OPEN", "OPEN", "OPEN", "O…
## $ tdd                 <int> 2, 2, 10, 15, 15, 32, 15, 38, 5, 5, 5, 5, 15, 2, …
## $ nCloc_affected_file <int> 248, 248, 248, 248, 248, 248, 248, 248, 248, 248, …
## $ complexity          <int> 32, 32, 32, 32, 32, 32, 32, 32, 32, 32, 32, 32, 32…
## $ origin              <chr> "PRE-EXISTING", "PRE-EXISTING", "PRE-EXISTING", "PR…
```

RQ1

TDD (Technical Debt Density) by PR Distribution

```
## Warning in geom_text(aes(x = 0, y = 0, label = paste("Mean:", round(mean(tdd), : All aesthetics have length 1, but the da
ta has 203 rows.
## ℹ Please consider using `annotate()` or provide this layer with data containing
##   a single row.
```

```
## Warning in geom_text(aes(x = 0, y = 0, label = paste("Median:", round(median(tdd), : All aesthetics have length 1, but th
e data has 203 rows.
## ℹ Please consider using `annotate()` or provide this layer with data containing
##   a single row.
```

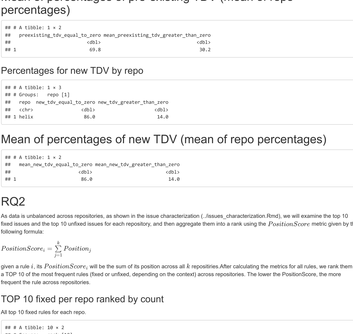


TDV (Technical Debt Variation)

TDV Distribution

```
## Warning in geom_text(aes(x = 0, y = 0, label = paste("Mean:", round(mean(tdv), : All aesthetics have length 1, but the da
ta has 203 rows.
## ℹ Please consider using `annotate()` or provide this layer with data containing
##   a single row.
```

```
## Warning in geom_text(aes(x = 0, y = 0, label = paste("Median:", round(median(tdv), : All aesthetics have length 1, but th
e data has 203 rows.
## ℹ Please consider using `annotate()` or provide this layer with data containing
##   a single row.
```



Percentages for TDV < 0 (TD decrease), TDV = 0 (TD unchanged) and TDV > 0 (TD increased) by repo

```
## # A tibble: 1 × 4
##   repo  tdv_lower_than_zero tdv_equal_to_zero tdv_greater_than_zero
##   <chr>               <dbl>              <dbl>                 <dbl>
## 1 helix                13.3               43.4                  43.4
```

Mean of percentages of TDV (mean of repo percentages)

```
## # A tibble: 1 × 3
##   mean_tdv_lower_than_zero mean_tdv_equal_to_zero mean_tdv_greater_than_zero
##                      <dbl>                  <dbl>                     <dbl>
## 1                     13.3                   43.4                      43.4
```

Percentages of pre-existing TDV by repo

```
## # A tibble: 1 × 3
##   repo  preexisting_tdv_equal_to_zero preexisting_tdv_greater_than_zero
##   <chr>                         <dbl>                             <dbl>
## 1 helix                          69.8                              30.2
```

Mean of percentages of pre-existing TDV (mean of repo percentages)

```
## # A tibble: 1 × 2
##   preexisting_tdv_equal_to_zero mean_preexisting_tdv_greater_than_zero
##                           <dbl>                                  <dbl>
## 1                          69.8                                   30.2
```

Percentages for new TDV by repo

```
## # A tibble: 1 × 3
## # Groups:   repo [1]
##   repo  new_tdv_equal_to_zero new_tdv_greater_than_zero
##   <chr>                 <dbl>                     <dbl>
## 1 helix                  86.0                      14.0
```

Mean of percentages of new TDV (mean of repo percentages)

```
## # A tibble: 1 × 2
##   mean_new_tdv_equal_to_zero mean_new_tdv_greater_than_zero
##                        <dbl>                          <dbl>
## 1                       86.0                           14.0
```

RQ2

As data is unbalanced across repositories, as shown in the issue characterization (../issues_characterization.Rmd), we will examine the top 10 fixed issues and the top 10 unfixed issues for each repository, and then aggregate them into a rank using the $PositionScore$ metric given by the following formula:

$$PositionScore_i = \sum_{j=1}^{k} Position_j$$

given a rule i, its $PositionScore_i$ will be the sum of its position across all k repositories.After calculating the metrics for all rules, we rank them in a TOP 10 of the most frequent rules (fixed or unfixed, depending on the context) across repositories. The lower the PositionScore, the more frequent the rule across repositories.

TOP 10 fixed per repo ranked by count

All top 10 fixed rules for each repo.

```
## # A tibble: 10 × 2
## # Groups:   rank [10]
##    rank helix
##   <int> <chr>
##  1     1 java:S1192  - 74
##  2     2 java:S106   - 64
##  3     3 java:S116   - 51
##  4     4 java:S1075  - 44
##  5     5 java:S125   - 32
##  6     6 java:S1874  - 26
##  7     7 java:S1135  - 18
##  8     8 java:S1170  - 16
##  9     9 java:S2864  - 10
## 10    10 java:S1481  - 10
```

TOP 10 fixed

TOP 10 fixed rules by PositionScore metric.

```
## # A tibble: 10 × 2
##    rule      position_score
##    <chr>              <dbl>
##  1 java:S1192           122
##  2 java:S106            123
##  3 java:S116            124
##  4 java:S1075           125
##  5 java:S125            126
##  6 java:S1874           127
##  7 java:S1135           128
##  8 java:S1170           129
##  9 java:S1481           130
## 10 java:S2142           131
```

TOP 10 unfixed per repo ranked by count

All top 10 unfixed rules for each repo.

```
## # A tibble: 10 × 2
## # Groups:   rank [10]
##    rank helix
##   <int> <chr>
##  1     1 java:S106   - 3022
##  2     2 java:S1192  - 2347
##  3     3 java:S116   - 1929
##  4     4 java:S1874  - 1056
##  5     5 java:S1186  - 749
##  6     6 java:S2293  - 572
##  7     7 java:S6213  - 502
##  8     8 java:S117   - 324
##  9     9 java:S125   - 300
## 10    10 java:S3008  - 276
```

TOP 10 unfixed

TOP 10 unfixed rules by PositionScore metric.

```
## # A tibble: 10 × 2
##    rule      position_score
##    <chr>              <dbl>
##  1 java:S106            122
##  2 java:S1192           123
##  3 java:S116            124
##  4 java:S1874           125
##  5 java:S1186           126
##  6 java:S2293           127
##  7 java:S6213           128
##  8 java:S117            129
##  9 java:S125            130
## 10 java:S3008           131
```

Outliers

Higher TDD

```
## # A tibble: 203 × 5
##    repo  pr_number total_debt total_ncloc   tdd
##    <chr>     <int>      <int>       <int> <dbl>
##  1 helix      2439        102          34  3
##  2 helix      2549        293         140  2.09
##  3 helix      1483        294         159  1.85
##  4 helix      2579        921         515  1.79
##  5 helix      2248        100          57  1.75
##  6 helix      2614       1003         687  1.65
##  7 helix      2505        230         141  1.63
##  8 helix      1943        202         128  1.58
##  9 helix      1492        118          75  1.57
## 10 helix      2281        914         585  1.56
## # ℹ 193 more rows
```

Higher TDV

```
## # A tibble: 203 × 3
##    pr_number repo    tdv
##        <int> <chr> <int>
##  1      2680 helix   368
##  2      1452 helix   343
##  3      2290 helix   332
##  4      2017 helix   285
##  5      2736 helix   226
##  6      1487 helix   206
##  7      2546 helix   179
##  8      2549 helix   147
##  9      2576 helix   144
## 10      2505 helix   133
## # ℹ 193 more rows
```

Lower TDV

```
## # A tibble: 203 × 3
##    pr_number repo    tdv
##        <int> <chr> <int>
##  1      1584 helix  -478
##  2      1593 helix  -476
##  3      1734 helix  -255
##  4      2235 helix  -209
##  5      2557 helix  -148
##  6      2505 helix  -135
##  7      1483 helix   -68
##  8      1592 helix   -50
##  9      1750 helix   -39
## 10      1760 helix   -30
## # ℹ 193 more rows
```

Extra

Number of PRs with pre-existing TD

```
##    n
## 1 192
```

Number of java:S1192 issues that affect test code

```
##    n
## 1 2418
```

Number of java:S1192 issues that affect not test code

```
##   n
## 1 3
```

Summary NCLOC

```
##      ncloc
##  Min.   :   3.0
##  1st Qu.: 190.5
##  Median :  442.0
##  Mean   :  863.6
##  3rd Qu.: 1052.0
##  Max.   :14176.0
```